Gait Sharing Toolkit- Tutorial

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Gait Sharing is a free, open-source toolkit designed to facilitate data management and decision-making in clinical gait analysis.

Mehrdad Davoudi received his master's degree in mechanical engineering from Sharif University of Technology in Iran and conducted his research in electromyographic signal processing for patients with cerebral palsy at Heidelberg University Hospital in Germany. He accepted a short-term research associate position at the Children's Hospital in Basel, Switzerland, during his transition phase to continue his journey in biomechanics.

```
USER_INFO = {
    "name":        "Mehrdad Davoudi",
    "title":       "Associate Researcher",
    "institution": "University Children's Hospital Basel (UKBB)",
    "location":    "Basel, Switzerland",
    "email":       "Mehrdaddavoodi15@yahoo.com, Mehrdad.davoudi@ukbb.ch",
    "LinkedIn":    "https://www.linkedin.com/in/mehrdad-davoudi-profile",
}
```

1. Installation and Running

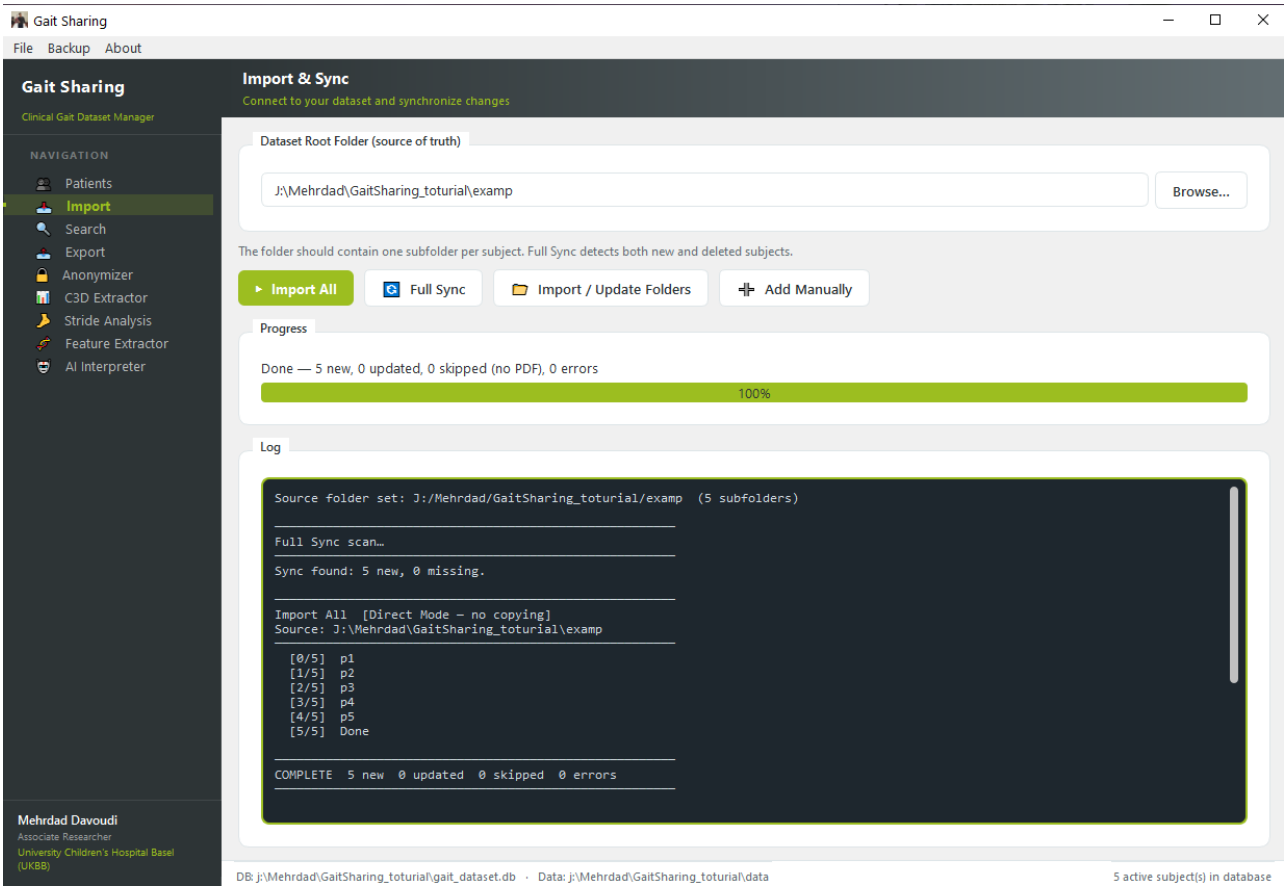
The toolkit has been developed with Python 3.13. Although Python 3.14 is available, the ezc3d library does not yet provide pre-built wheels for this version. Therefore, Python 3.13 is recommended to ensure full compatibility with all dependencies.

```bash
pip install -r requirements.txt
python GaitSharing_main.py
```

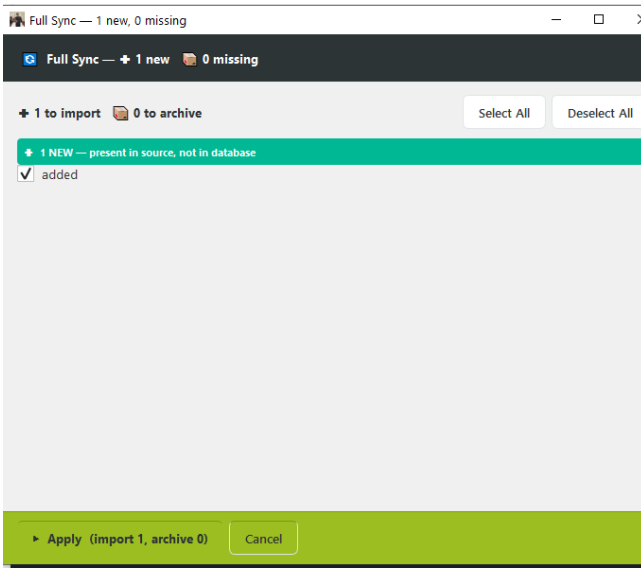
2. Import Window

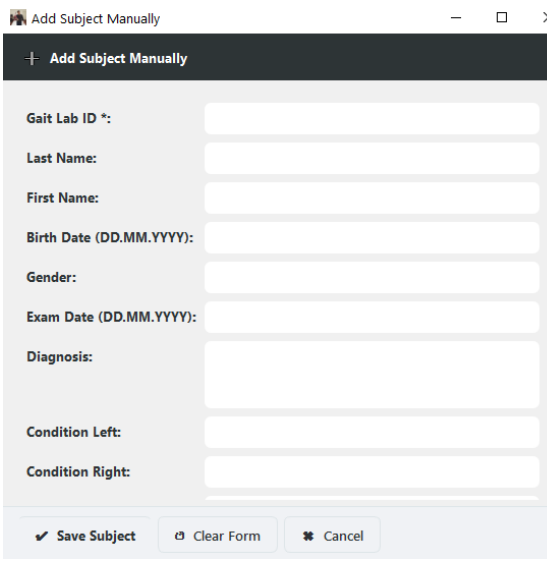
By navigating to the existing lab database (Browse), the toolkit reads the patient report files and automatically extracts their information to be transferred into the Patients window (Import All).

The existing database consists of folders named with the subjects' IDs (e.g., p1 to p5), and includes gait data such as C3D files containing kinematics and kinetics, along with other clinical data and often a PDF file with patient information (name, last name, birthdate, etc.). The toolkit depends on this PDF to extract detailed information and create the local database.



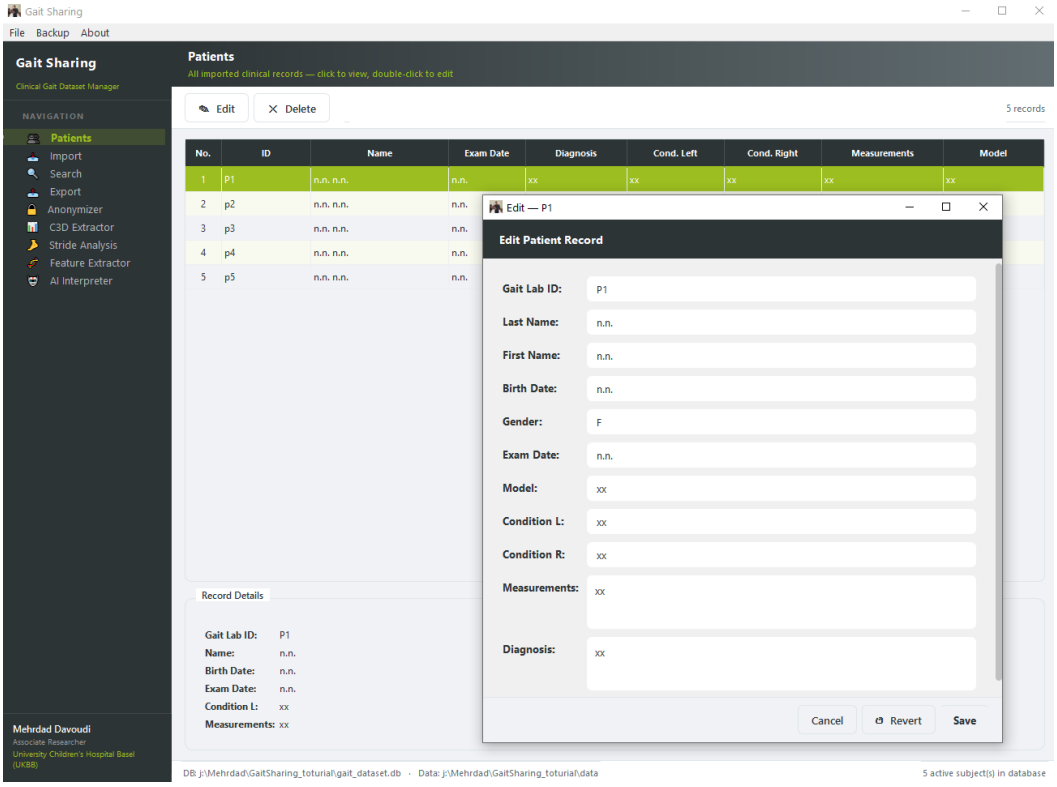
In case of changes in the original database (e.g., removing or adding documents or patients), pressing "Full Sync" will detect the changes and update the patients' information accordingly. Subjects can also be defined manually (Add Manually) or imported from a different database (Update Folder); however, they will all be managed through the same Patients window.





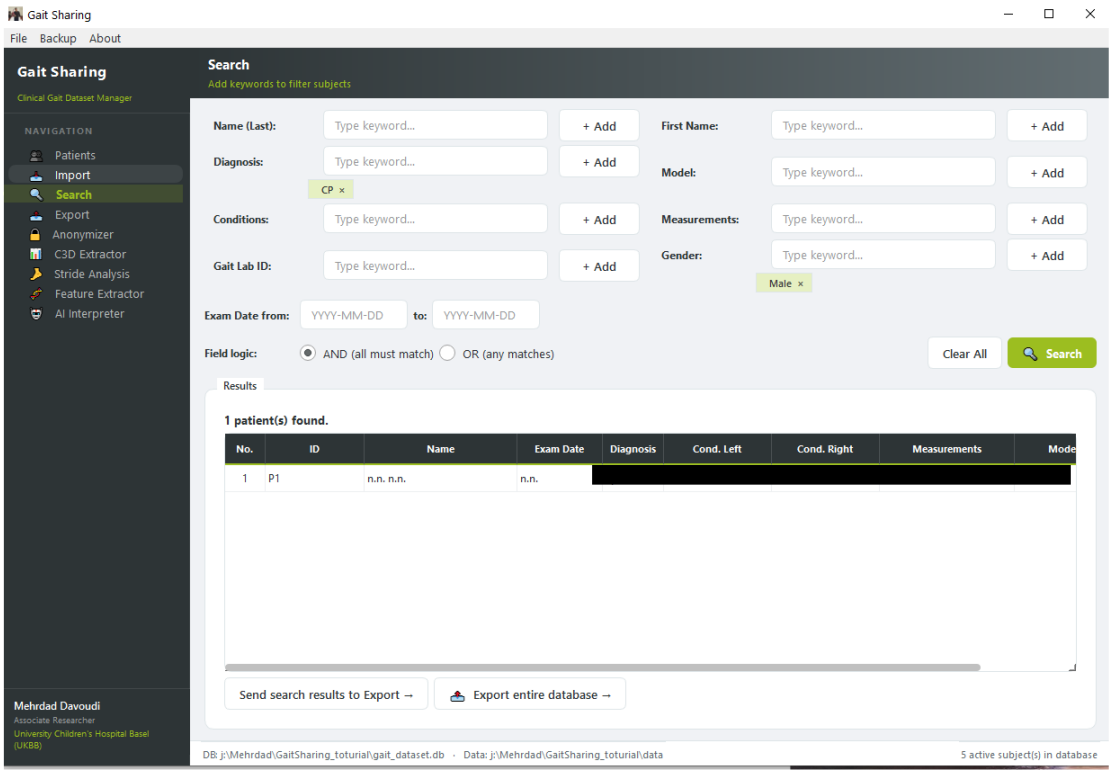
3. Patients Window

The imported patients' records, including name, last name, diagnosis, examination date, etc., can be viewed in this window. Records can be edited or deleted by double-clicking on them.



4. Search Window

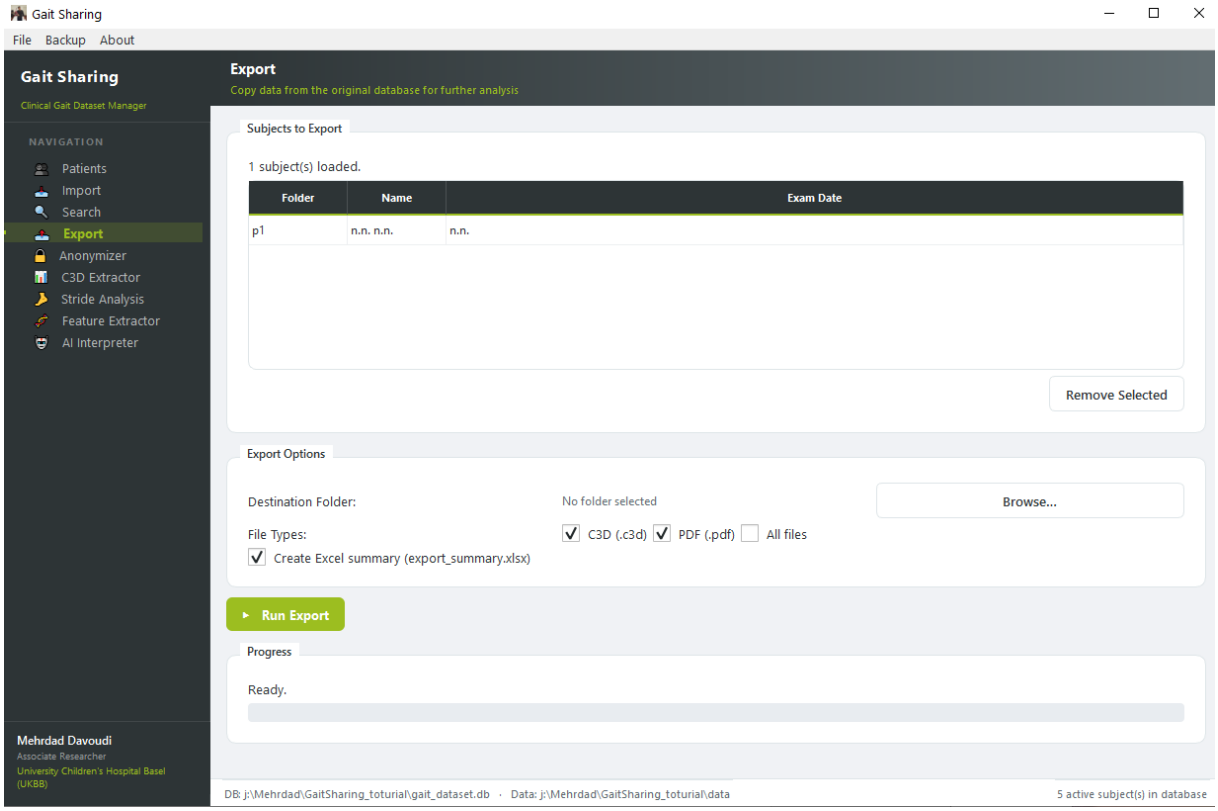
Patients can be filtered by their clinical information, gender, and examination date. The search logic can be set to AND (all conditions must be satisfied) or OR (at least one condition satisfied).



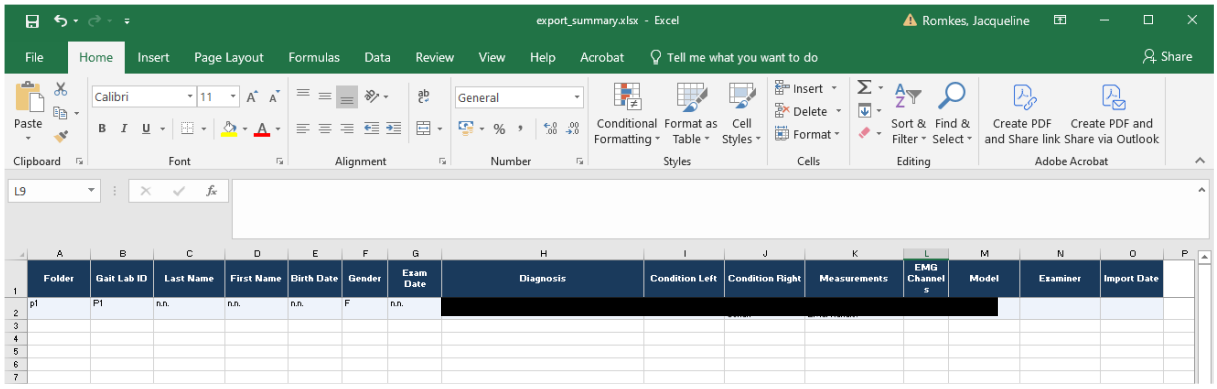
The list of filtered subjects (or the entire database) can be exported to the next window for further analysis. This provides an overview of the existing subjects and is essential for managing large databases.

5. Export Window

This option helps create a secondary database containing only specific files (e.g., C3D for research purposes) for the filtered patients. This secondary database avoids applying changes to the original patient data, especially when planning further processing. The location for this collection can be selected through the Browse button.

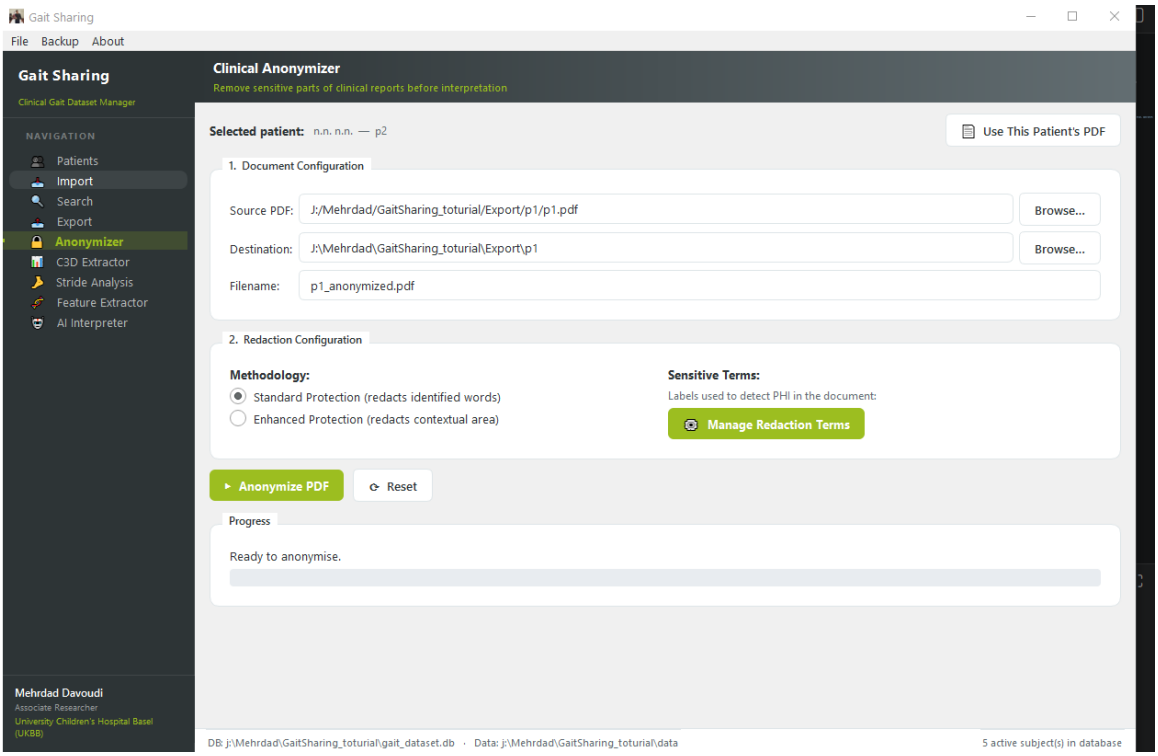


A summary Excel file (export_summary.xlsx) will also be generated during the export process.

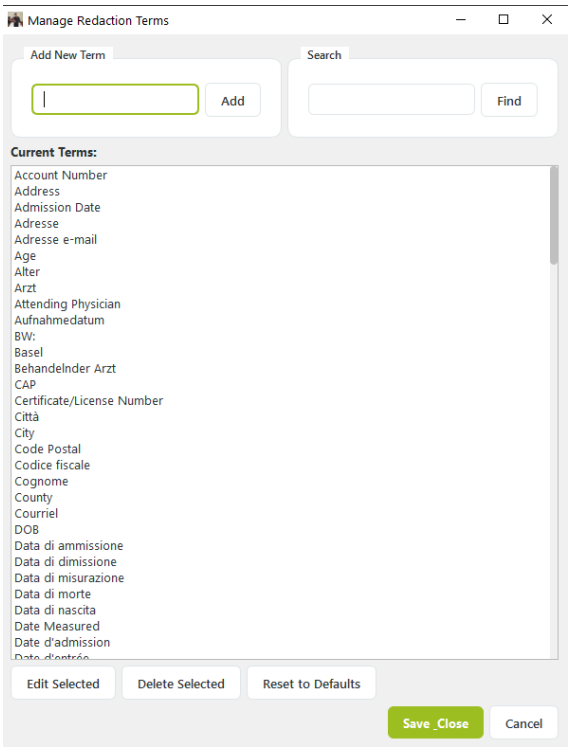


6. Anonymizer

To share clinical outputs (e.g., with generic LLMs or other centers), patient data must first be anonymized, especially the patient reports. After selecting the file, the location and name of the anonymized file can be specified.

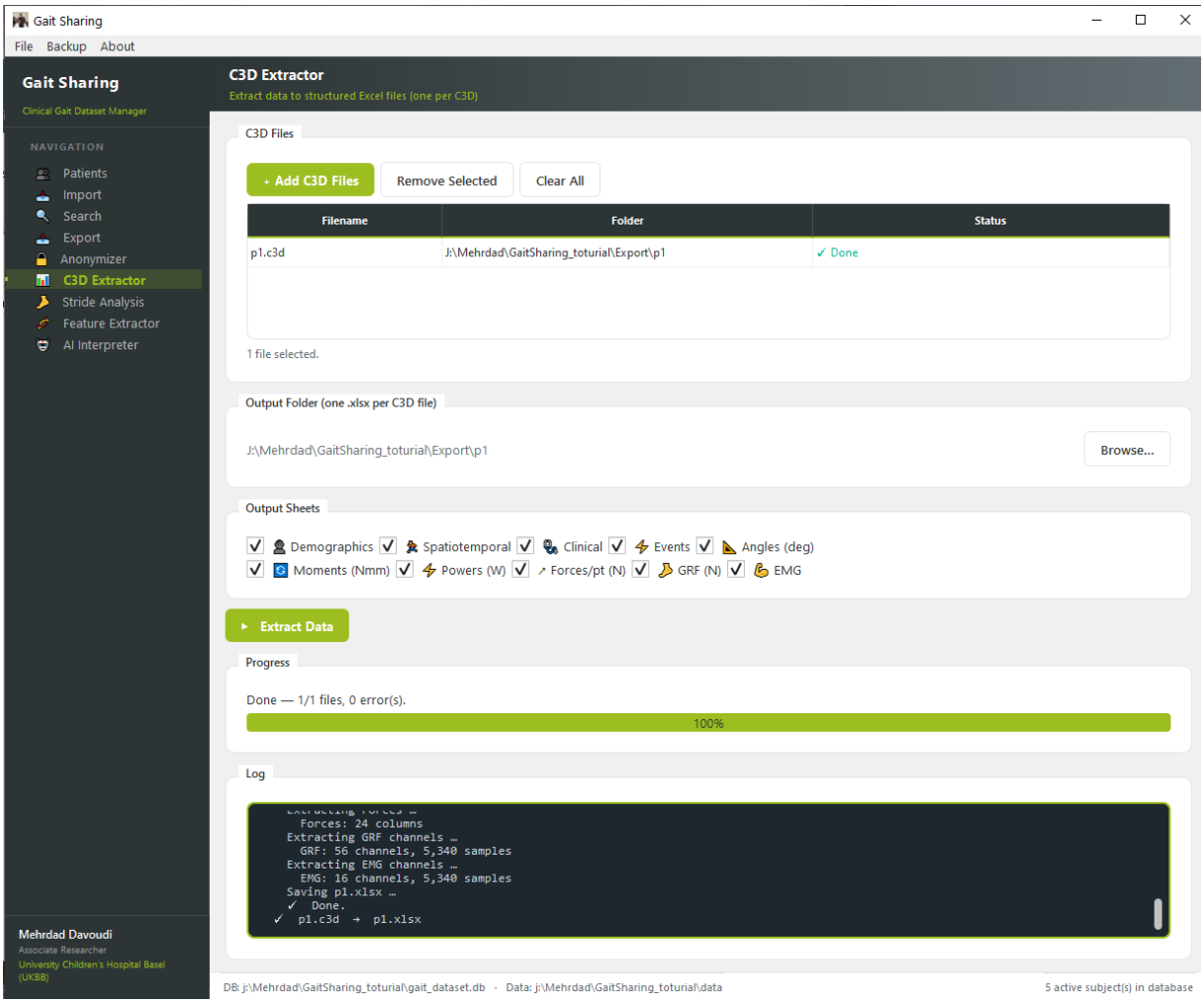


This process works by extracting text from PDFs and searching for sensitive keywords from a predefined list. The list is provided in German, French, Italian, and English. Sensitive terms can be added or edited by clicking the Manage Redaction Terms button. There are two options for the redaction process: Standard, which covers only the exact detected word in the list, or Enhanced, which aggressively covers the term and area around it (recommended for clinical purposes).



7. C3D Extractor

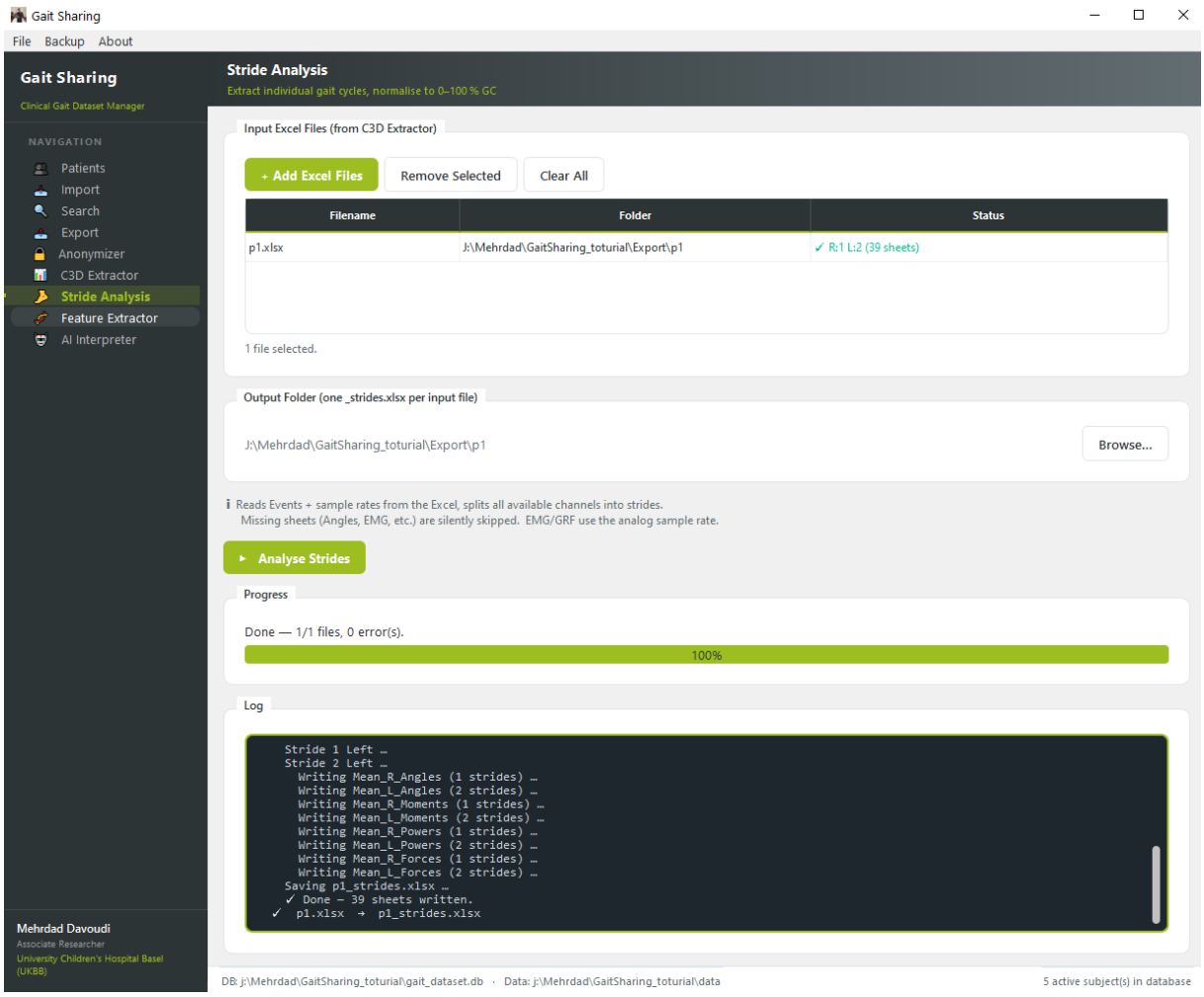
C3D is the standard format for saving the outputs of clinical gait analysis. However, working with this type of file is challenging, especially for clinicians. The C3D Extractor provides the ability to extract the required data from one or multiple C3D files into an Excel spreadsheet.

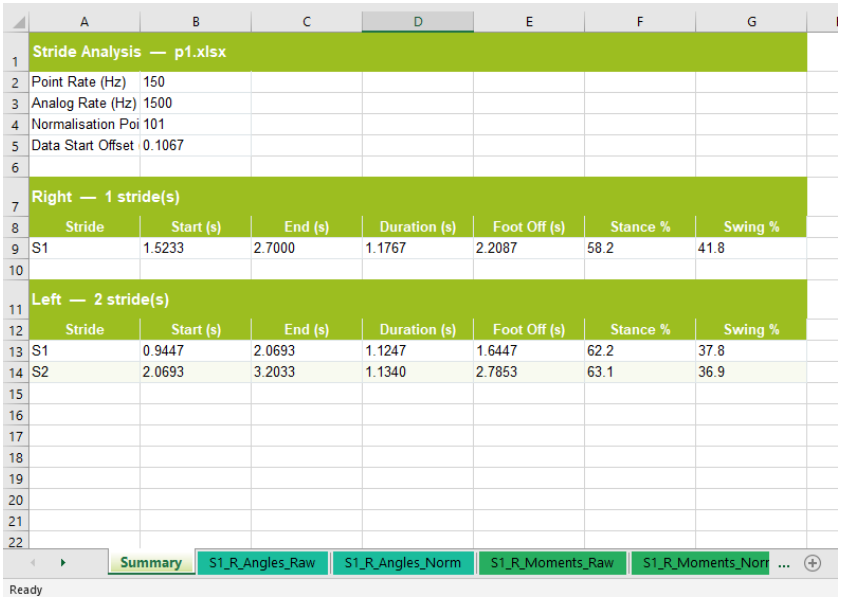


The output (.xlsx format) contains the results for the entire gait cycle, extracted from the data saved in the C3D file.

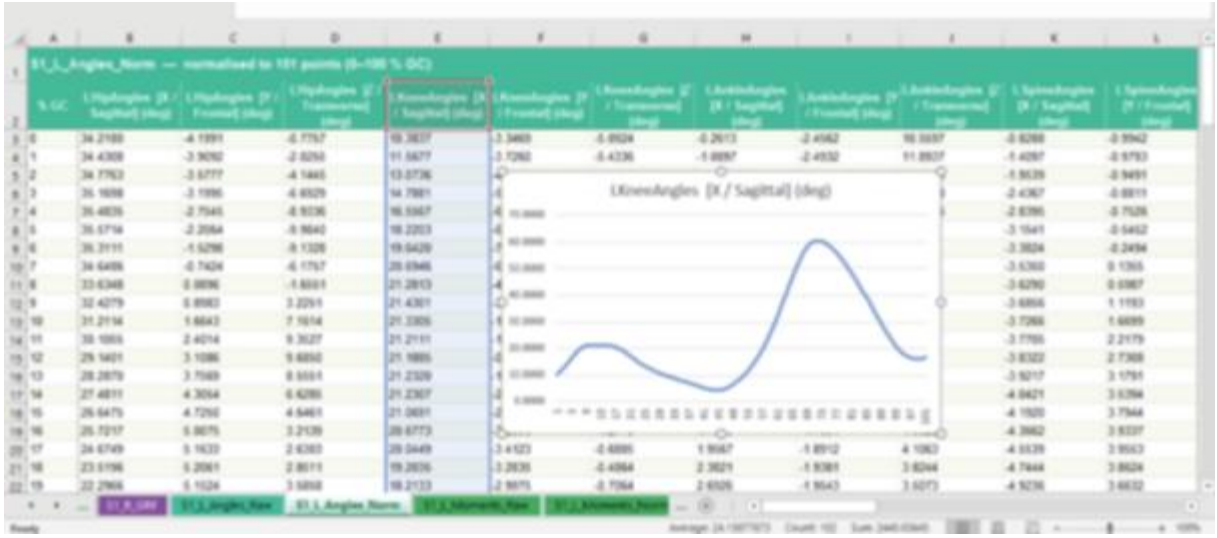
8. Stride Analysis

The output from the previous step in Excel format serves as the input for stride segmentation. To extract the strides, in addition to the timing of events (foot off and heel contact), the start frame of data capturing (due to cropping) will be considered. This information must be available inside the C3D file. A C3D file can contain several strides recorded for each side (right or left), with multiple variables of kinematics and kinetics, and they will be saved in different sheets of a single Excel file.





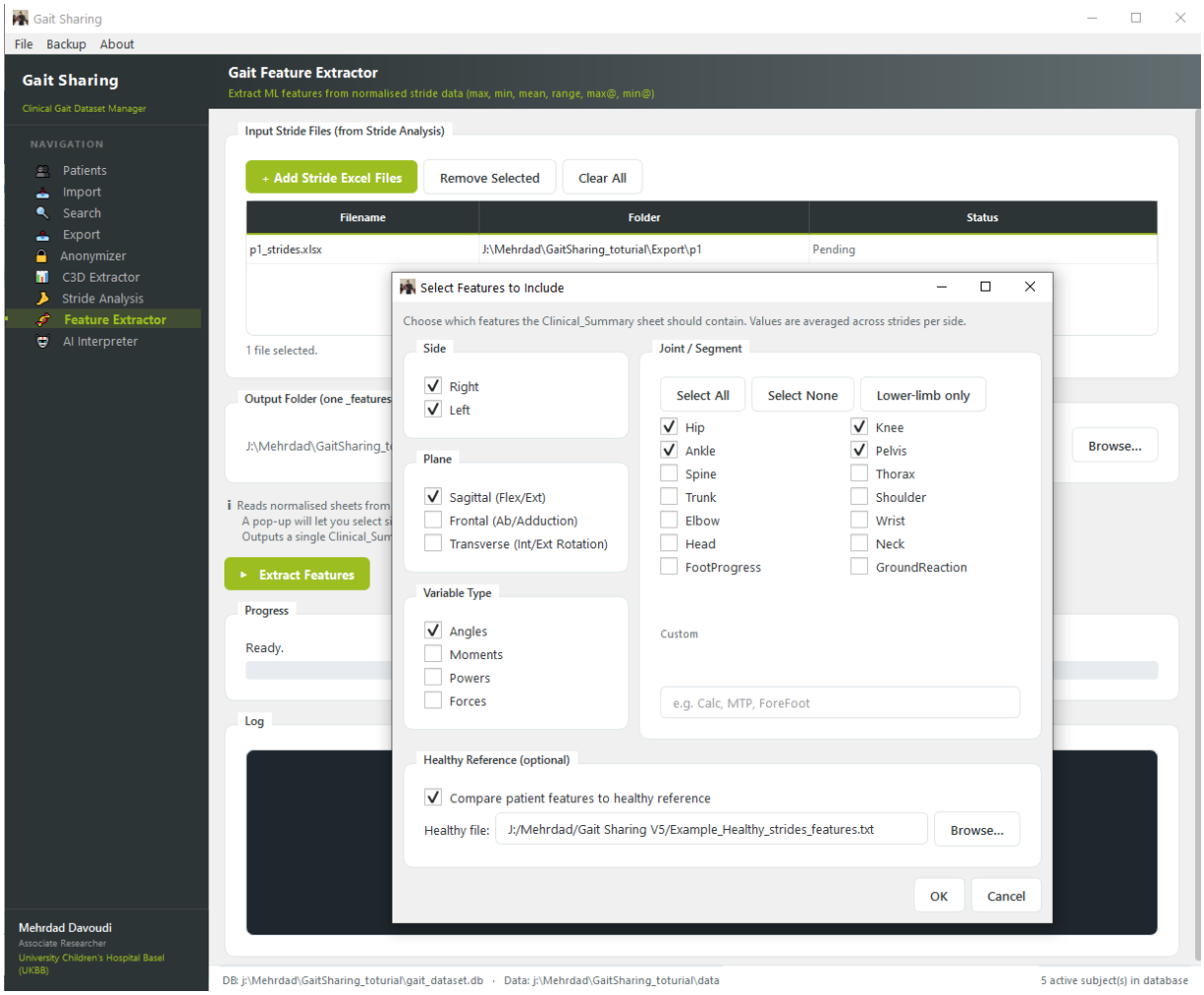
	A	B	C	D	E	F	G
1	Stride Analysis — p1.xlsx						
2	Point Rate (Hz)	150					
3	Analog Rate (Hz)	1500					
4	Normalisation Poi	101					
5	Data Start Offset	0.1067					
6							
7	Right — 1 stride(s)						
8	Stride	Start (s)	End (s)	Duration (s)	Foot Off (s)	Stance %	Swing %
9	S1	1.5233	2.7000	1.1767	2.2087	58.2	41.8
10							
11	Left — 2 stride(s)						
12	Stride	Start (s)	End (s)	Duration (s)	Foot Off (s)	Stance %	Swing %
13	S1	0.9447	2.0693	1.1247	1.6447	62.2	37.8
14	S2	2.0693	3.2033	1.1340	2.7853	63.1	36.9



9. Feature Extraction

As I have discussed on my LinkedIn, ChatGPT is not able to produce accurate interpretations when given numerous curves as input. Rearranging the information into a tabular format yields significantly better outputs. Therefore, a Feature Extraction layer has been implemented within the toolkit. The extracted features include min, max, mean, range, min@ (time of gait cycle), and max@ (time of gait cycle) for the whole gait cycle, stance phase, and swing phase, computed for each stride and side (right and left).

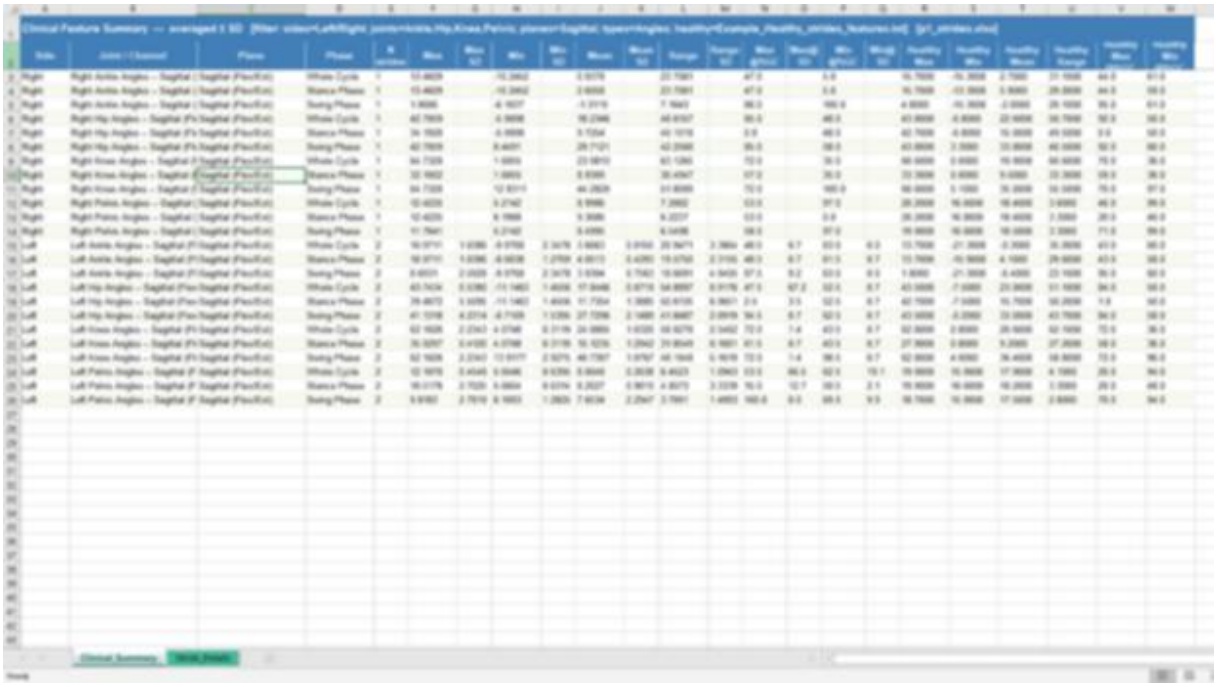
The input for this stage is the segmented strides saved in the previous step.



In addition, the toolkit allows the user to select which variables are important for interpretation; calculations will be performed only on those selected variables.

One important limitation of LLMs like ChatGPT is that there are no normal (healthy) reference values embedded in their training data. To address this, the toolkit provides the option for users to create normal reference values through the previous pipelines and upload them here. By enabling the "Compare with reference data" option, the outputs will include both patient and healthy data, enabling direct comparison.

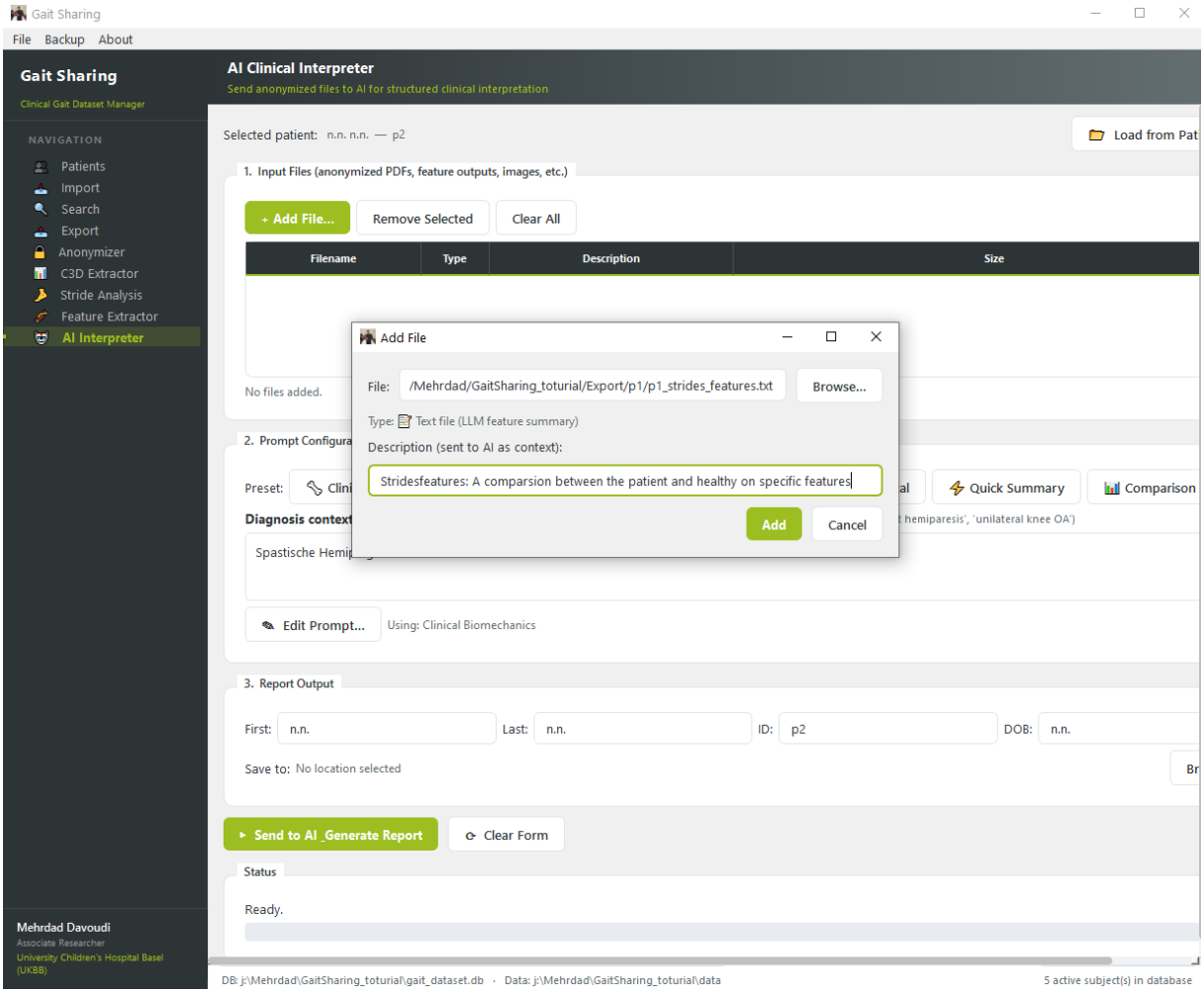
The results are available in both .xlsx and .txt formats. The text file can be used directly in the AI Interpretation step. In the Excel file, there is a sheet for features at the stride level and also the average (and SD) across strides when more than one stride is available.



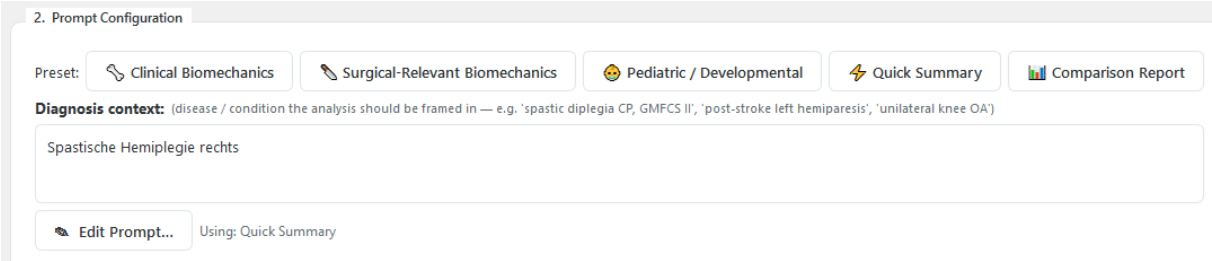
10. AI Interpreter

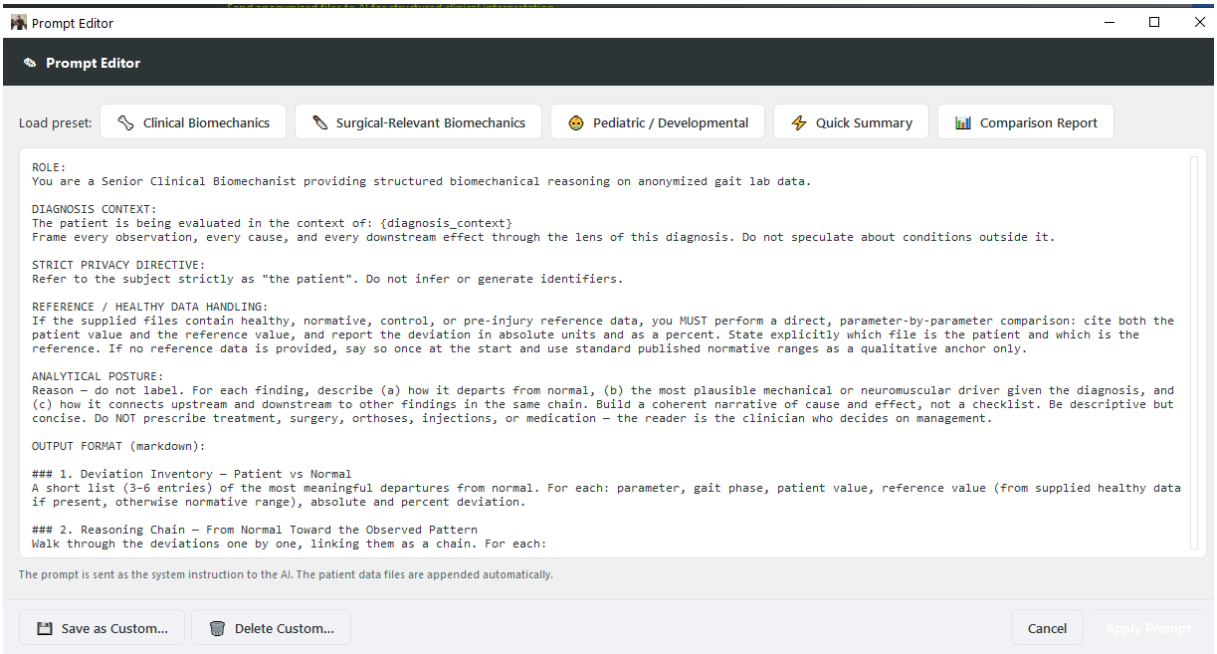
This part of the toolkit is its core feature, providing the ability to communicate with LLMs (currently ChatGPT) and request a description of the measured data.

First, the user selects which files to transfer. These can include the features.txt file to convey the deviations and the anonymized PDF file for additional information that may not be available in the features, such as spatiotemporal parameters. The user will also be asked to provide a description of the uploaded file.



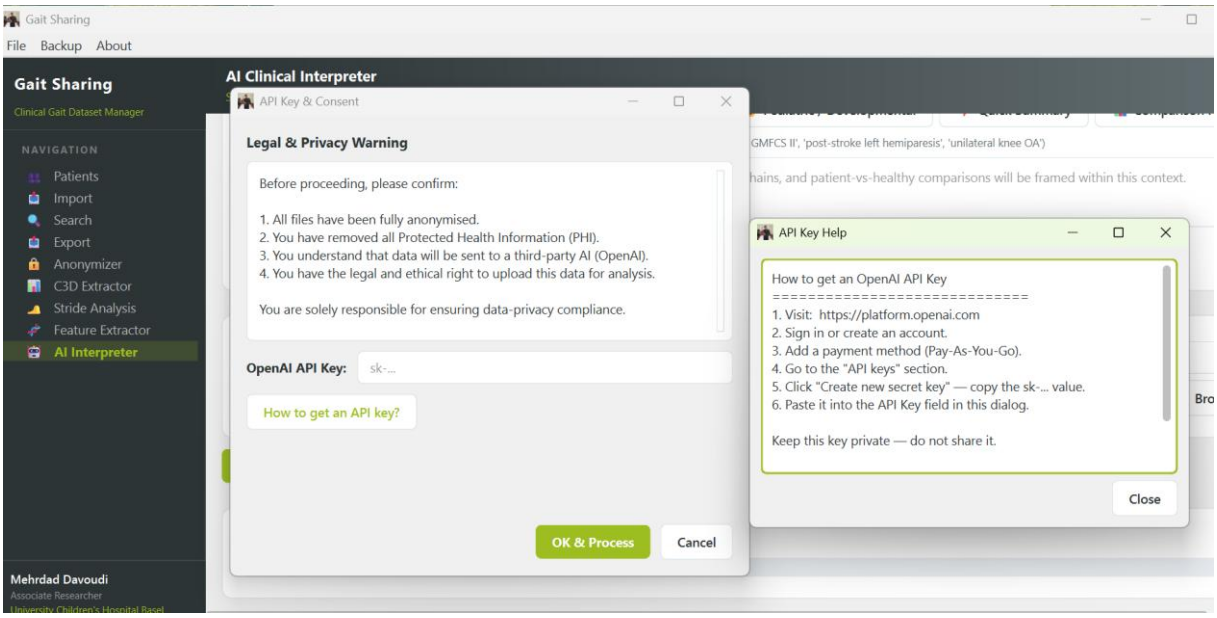
The prompts sent to ChatGPT can be engineered within the tool. There are already five different prompt categories available, and prompts can be customized and saved for future use.





The Diagnosis context (e.g., cerebral palsy) should be entered in the designated box to guide the LLM in generating a description of the patient's gait. The output report will be named after the patient; however, this information remains local and is not transmitted to OpenAI.

To access OpenAI, the user needs a personal API key. Instructions for obtaining this key are provided in a pop-up window. In case of difficulty acquiring the key, please contact me for support.



➢ Output

To evaluate the performance of the toolkit, a set of synthetic gait features (manually created data) with a format like the output of the Feature Extraction step, including spatiotemporal parameters (e.g., speed) and specific clinical data (e.g., strength), was used for the AI Interpretation stage. In the diagnosis box, "cerebral palsy, unilateral, with right side affected" was entered to evaluate the system in the presence of pathology. The data was manipulated to simulate an equinus gait condition to test whether it could be recognized by the toolkit; this condition was not mentioned in the diagnosis.

The following shows how the Quick Summary Prompt option performed in describing the input. The numerical values were replaced with "X". However, by comparing the outputs with the input data, the performance in describing the input variables was accurate.

I would appreciate it if you tested the other embedded prompts on your real cases and share your feedback with me.

As mentioned in the application, the responsibility for legal compliance rests with the user. However, uploading anonymized features —> not curves, which could serve as biomarkers <— along with a user-provided summarized diagnosis to guide the LLM, and receiving a detailed description of the measurements and general cause-and-effect chain, represents the safest approach through this platform.

Quick Summary Prompt Output

Rapid Reasoning Summary

1. Pattern: Right spastic hemiplegic gait with mild equinus at initial contact plus right in-toeing.

2. Lead deviation: Right in-toeing. The patient's right foot progression angle is internally rotated on average (−X°) versus the left (+X°), a ~X° asymmetry; relative to typical external foot progression of ~X–X°, this is ≈X–X° more internal than expected. Within right spastic hemiplegia, this is most plausibly driven by reduced external tibial torsion on the right (transmalleolar axis X° right vs X° left) together with dynamic internal hip rotation (right hip rotation mean −X°).

3. Downstream chain: The internal foot progression shortens the effective foot lever in late stance, reducing push-off efficiency and contributing to shorter steps and slower walking. Concurrent mild equinus/forefoot initial contact on the right (ankle −X° plantarflexed at X% GC; normative heel-strike near neutral, quantitative lab reference not provided) aligns with impaired dorsiflexion, further promoting cautious loading and longer double support.

4. Asymmetry: Present; right worse. Evidence: right stance X% (≈−X% vs X% reference) vs left X% (≈+X% vs reference), and right in-toeing vs left out-toeing as above. These findings are consistent with the right being the hemiplegic side.

5. Functional consequence: The patient walks slower and more cautiously than healthy controls (speed X vs X m/s, −X%; stride length X vs X m, −X%; cadence X vs X steps/min, −X%; stride width X vs X m, +X%; double support X% vs X%, +X%), indicating increased energetic cost and a stability-seeking strategy, with potential elevated trip risk on the right from plantarflexed initial contact and reduced swing-phase dorsiflexion.